

Chi-Square Distribution

If a random variable X has a chi-square distribution with n d.f., we write $X \sim \chi^2_{(n)}$ and its p.d.f. is given by :

$$f(x) = \frac{1}{2^{n/2} \Gamma(n/2)} e^{-x/2} x^{(n/2) - 1} ; 0 \leq x < \infty$$

Properties of χ^2 -Distribution

1. A rough sketch of the probability curve of the χ^2 -distribution for $\nu = 3$ and $\nu = 6$ is given in Fig. 8.4.
2. As ν becomes smaller and smaller, the curve is skewed more and more to the right. As ν increases, the curve becomes more and more symmetrical.
3. The mean and variance of the χ^2 -distribution are ν and 2ν respectively.

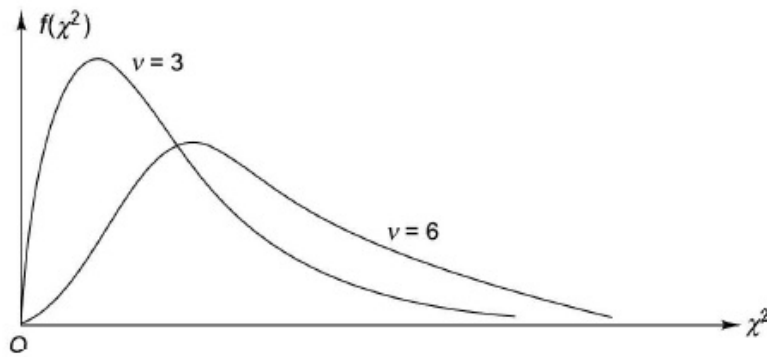


Fig. 8.4

4. As ν tends to ∞ , the χ^2 -distribution becomes a normal distribution.

Uses of χ^2 -Distribution

1. χ^2 -distribution is used to test the goodness of fit. i.e., it is used to judge whether a given sample may be reasonably regarded as a simple sample from a certain hypothetical population.
2. It is used to test the independence of attributes. That is, if a population is known to have two attributes (or traits), then χ^2 -distribution is used to test whether the two attributes are associated or independent, based on a sample drawn from the population.

χ^2 -Test of Goodness of Fit

The Chi-square goodness of fit test is a statistical hypothesis test used to determine whether a variable is likely to come from a specified distribution or not. It is often used to evaluate whether sample data is representative of the full population.

If O_i ($i = 1, 2, \dots, n$) is a set of observed (experimental) frequencies and E_i ($i = 1, 2, \dots, n$) is the corresponding set of expected (theoretical or hypothetical) frequencies, then χ^2 is defined as

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right]$$

where $\sum O_i = \sum E_i = N$ (total frequency) and degrees of freedom ($d.f.$) = $(n - 1)$.

Conditions for Applying the χ^2 Test

Following are the conditions that should be satisfied before the χ^2 test can be applied.

(a) N , the total number of frequencies, should be large. It is difficult to say what constitutes largeness, but as an arbitrary figure, we may say that **N should be at least 50**, however few the cells.

(b) No theoretical cell-frequency should be small. Here again, it is difficult to say what constitutes smallness, but 5 should be regarded as the very minimum and **10 is better**. If small theoretical frequencies occur (*i.e.*, < 10), the difficulty is overcome by grouping two or more classes together before calculating $(O - E)$. **It is important to remember that the number of degrees of freedom is determined with the number of classes after regrouping.**

(c) The constraints on the cell frequencies, if any, should be linear.

Pb : 1

The following table gives the number of accidents that took place in an industry during various days of the week. Test whether accidents are uniformly distributed over the week.

Day	Mon	Tue	Wed	Thu	Fri	Sat
No. of accidents	14	18	12	11	15	14

Soln:-

H_0 : Accidents occur uniformly over the week.

H_1 : Accidents does not occur uniformly over the week.

Total number of accidents = 90

Expected number of of accidents on any day = $\frac{90}{6} = 15$

$$\begin{array}{lcl} \therefore O_i & : & 14 \quad 18 \quad 12 \quad 11 \quad 15 \quad 14 \\ E_i & : & 15 \quad 15 \quad 15 \quad 15 \quad 15 \quad 15 \end{array}$$

$$\begin{aligned} \chi^2 &= \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = \frac{(14-15)^2}{15} + \frac{(18-15)^2}{15} + \frac{(12-15)^2}{15} + \frac{(11-15)^2}{15} + \frac{(15-15)^2}{15} + \frac{(14-15)^2}{15} \\ &= \frac{1}{15} + \frac{9}{15} + \frac{9}{15} + \frac{16}{15} + 0 + \frac{1}{15} \\ &= 2 \end{aligned}$$

degrees of freedom $\delta = n - 1 = 6 - 1 = 5$.

From the χ^2 -table $\chi_{0.05}^2 = 11.070$

Since $\chi^2 < \chi_{0.05}^2$, we accept the null hypothesis.

A die is thrown 276 times and the results of these throws are given below:

No. appeared on the die	1	2	3	4	5	6
Frequency	40	32	29	59	57	59

Test whether the die is biased or not.

Soln:- H_0 : The die is unbiased
 H_1 : The die is biased.

Total Frequency = 276

Expected frequency on each number on the die = $\frac{276}{6} = 46$.

Now, O_i : 40 32 29 59 57 59
 E_i : 46 46 46 46 46 46.

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i} = \frac{1}{46} [6^2 + 4^2 + 17^2 + 13^2 + 11^2 + 13^2]$$
$$= 21.30$$

From the χ^2 table $\chi^2_{0.05} = 11.070$

Since $\chi^2 > \chi^2_{0.05}$, we reject the null Hypothesis.

$\therefore$ The die is biased.

The following table shows the distribution of digits in numbers chosen at random from a telephone directory:

Digits	0	1	2	3	4	5	6	7	8	9
Frequency	1026	1107	997	966	1075	933	1107	972	964	853

Test whether the digits may be taken to occur equally frequently in the directory.

Soln:- H_0 : The digits taken in the directory occur with equal frequency.

H_1 : The digits taken in the directory occur with unequal frequency.

Total Frequency = 10,000.

Expected number of frequency for each digit is $\frac{10000}{10} = 1000$

Now,

O_i	1026	1107	997	966	1075	933	1107	972	964	853
E_i	1000	1000	1000	1000	1000	1000	1000	1000	1000	1000

$$\chi^2 = \frac{1}{1000} \left[(26)^2 + (107)^2 + (3)^2 + (34)^2 + (75)^2 + (67)^2 + (107)^2 + (28)^2 + (36)^2 + (141)^2 \right]$$

$$= 58.54^2$$

degrees of freedom is $\delta = 10 - 1 = 9$.

From χ^2 table $\chi^2_{0.05} = 16.919$

Since $\chi^2 > \chi^2_{0.05}$, we reject the null hypothesis.

Records taken of the number of male and female births in 800 families having four children are as follows:

No. of male births	0	1	2	3	4
No. of female births	4	3	2	1	0
No. of families	32	178	290	236	94

Test whether the data are consistent with the hypothesis that the binomial law holds and the chance of male birth is equal to that of female birth, namely $p = q = 1/2$.

Soln:- H_0 : The data are consistent with the hypothesis of equal probability for male and female births, (i.e.) $p = q = 1/2$.

H_1 : $p \neq q$.

We use binomial distribution to calculate the Expected frequency by

$$N(r) = N \times P(X=r)$$

where N is the total frequency, $N(r)$ is the number of families with r -male children.

$$E_1 = N(0) = 800 \times {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = 50$$

$$E_2 = N(1) = 800 \times {}^4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = 200$$

$$E_3 = N(2) = 800 \times {}^4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 300$$

$$E_4 = N(3) = 800 \times {}^4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 = 200$$

$$E_5 = N(4) = 800 \times {}^4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 = 50$$

Now

$$O_i: \quad 32 \quad 178 \quad 290 \quad 236 \quad 94$$

$$E_i: \quad 50 \quad 200 \quad 300 \quad 200 \quad 50$$

$$\chi^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 54.433$$

degree of freedom = 4
 $\chi^2_{0.05} = 9.488$

Since $\chi^2 > \chi^2_{0.05}$, we reject the null hypothesis.